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M-205 DEAERATOR

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with **Part 4, Subsection 1.2. “Summary of Work”**.

The bidder shall supply One (1) Deaerator and Storage Tank for each block **or one (1) deaerator and storage tank for each HRSG.**

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The deaerator and storage tank shall include, but not be limited to the following:

- Deaerating heater
- Storage tank
- Trays
- Spray valves
- Baffles
- Vent condensers
- Armored Gauge glasses
- Safety valves
- Vent orifices
- Painting
- Instrumentation & Control Equipment

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5**. The codes and standards of the following organizations extracted from **Part 4, Subsection 2.5 “Codes and Standards”** or international equivalent standards are specifically applicable to the design, manufacture and testing of the equipment included in this specification:

ANSI	American National Standards Institute.
ASME	American Society of Mechanical Engineers.
ASTM	American Society of Testing and Materials.

BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HEI	Heat Exchange Institute
JIS	Japan Industrial Standard

3. DESIGN AND OPERATING CONDITIONS

The deaerator shall be designed in accordance with the requirements of ASME Boiler and Pressure Vessel Code Section VIII, Division 1, and the Heat Exchange Institute standards.

The deaerator shall be designed for turbine-generator valves wide open (V.W.O) load corresponding to peak Combined Cycle Plant output. The deaerator shall be capable of continuous and safe operation over the full range of steam and water temperatures and pressures.

The deaerator shall also be designed to withstand safely and without damage full vacuum conditions.

Design pressure of heater and storage tank shall be equal to the maximum steam pressure in the deaerator plus a margin of 25%.

The deaerator shall be suitable for plant operation with either constant or sliding boiler pressure.

Guaranteed oxygen content over the whole load range of the combined cycle plant shall not be greater than 0.015 mg/l

4. DETAILED REQUIREMENTS

4.1 Deaerator Type

The Deaerator shall be spray-type, tray-type or spray-tray type.

4.2 Storage Tank

The minimum hot water storage capacity below normal operating level shall be not less than the equivalent of ten (10) minutes of feed-water flow at peak

combined cycle plant operation volume at normal water level shall not exceed 2/3 the total deaerator volume.

4.3 Shells and Heads

All shells and heads shall be constructed of carbon steel plates conforming to SA-515. Grade 70

4.4 Manholes

A circular manhole not smaller than 500 mm shall be provided for access to the heater section.

A circular manhole not less than 500 mm inside diameter shall be provided for access to the water storage section.

Manholes shall be of steel construction equipped with permanent gaskets. Access doors shall be hinged or equipped with davit assembly.

4.5 Trays and Other Internals

The deaerator trays shall be of ample area and shall be arranged to completely separate the air and non condensable gases from the incoming condensate. The trays shall be of such a design that they can be handled by one man. Trays shall be securely bolted in position such that they will not be dislodged on severe load changes or upsets in the deaerator.

Deaerator heater trays, spray pipes, spray valves and all internal components in contact with non-condensable gases shall be stainless steel.

4.6 Supports

The deaerating heater shall be provided with supports capable of supporting the entire weight of the unit (heater and storage tank) with all spaces flooded. One support shall be fixed by bolting to the structure and the other supports shall be sliding type. Saddle type supports shall be provided.

4.7 Nozzles

All nozzles shall be compatible with the requirements of **Part 4, Section 3, M-001 Piping**.

Nozzle connections on heater section and storage tank shall be suitably reinforced, per ASME, Division 1.

Carbon Steel nozzles in sizes 60 mm and larger shall be butt-welding with end preparation conforming to Part 4, Section 3, Specification M-001.

Socket-welding ends shall be provided in sizes 50 mm and smaller. Flanges, where required, shall be ANSI Standard, weld neck, forged steel, and pressure class suitable for the design conditions.

External Piping (if any): External piping shall consist of ASME SA-106, Grade B seamless carbon steel pipe, and steel fittings with butt-welded and socket welded joints for sizes as above specified for nozzles. Vent piping shall be 316L stainless Steel. All pipe 50 mm and smaller in size shall be Schedule 80 wall thickness.

All inlet nozzles shall be provided with 316L stainless steel impingement plates, except boiler feed pump recirculation lines shall be provided with spargers located below the minimum water level.

4.8 Boiler Feed Pump Suction Outlet

Each pump suction outlet shall be provided with a weir of sufficient height to prevent foreign objects or sludge from entering the pump Suction piping.

A grid-type strainer shall be provided over each pump suction outlet. Strainer shall be of heavy duty stainless steel construction and shall be firmly attached.

Each pump suction outlet shall be provided with vortex-eliminating baffles.

4.9 Vent Condenser

Vent condenser can be supplied and may be of the internal type, mounted within the heater shell and arranged to properly condense the condensable portion of the vent mixture and vent the non-condensable gases to the atmosphere. Appropriate vent orifices with bypass shall be furnished as required.

All internals of the vent condenser shall be of 316L stainless steel.

4.10 Safety Valves

Safety valves shall be provided to protect the deaerator and deaerator storage tank from overpressure from any source. The Bidder shall purpose the number of safety valves to be installed.

4.11 Painting

All parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and the exterior surfaces shall be painted in accordance with **Part 4, Section 3, S-420 and S-430**.

Before shipment, all exposed machined surfaces shall be coated with a suitable grease or antirust compound, and inlets and outlets shall be blanked off.

4.12 Nitrogen Sealing

All nozzles shall be blanked-off with special nitrogen-pressure tight gasketed shipping covers.

While under vacuum, nitrogen shall be introduced into the deaerator and brought up to the positive pressure recommended for shipment.

A pressure gauge with isolation valve shall be provided to monitor the nitrogen blanket. The Bidder shall maintain the nitrogen pressure to required level.

5. INSPECTION AND TESTING

Inspection and testing shall be per ASME VIII, Division I.

The vessel shall be subject to spot radiograph per ASME VIII, Division VI

A site test shall be conducted by the Bidder to demonstrate dissolved oxygen guarantees in accordance with the ASME Power Test Code for Deaerators, PTC 12.3.

The basis for determining oxygen guarantee shall be per ASTM D888 using Referee Method A (Colorimetric Indigo Carmine), or other similar method acceptable to the Owner. All deficiencies shall be corrected by the Bidder at no cost to the Owner.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance/control shall conform to the requirements of **Part 4, Subsection 2.7.**

7. INSTRUMENTATION AND CONTROLS

7.1 Gauge Glasses

Armored gauge glasses shall be provided by Bidder. Gauge glasses shall be of such total length as to effectively cover the entire storage tank.

7.2 Instrument and Standpipe Connections

The Bidder shall furnish and install suitable connections in accordance with Clause 4.7 herein for the pressure instruments and level instrument stand pipes including but not limited to the following:

- 50 mm valved connections at the top and bottom of the storage tank at each end of the tank for level instrument standpipes. Four (4) connections total.
- 15 mm valved connections for storage tank and deaerating section pressure indicators and pressure transmitters. Four (4) connections total.

7.3 Level Instrumentation

The Bidder shall provide deaerator level instruments to meet the requirements of **Part 4, Subsection 1.2.**

7.4 Temperature Measurement

The Bidder shall furnish and install a temperature indicator complete with thermowell in the storage tank.

Refer to Part 4, Section 3 I-001 “Instrumentation and Control” for the design requirement for Instrumentation and Control.

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